

Serum Routing Tool

Thermo Fisher Scientific

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ISE494: Senior Capstone Project

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Company Background

Thermo Fisher Scientific is a global medical supply company formed in 2006 from the union of Thermo Electron Corporation and Fisher Scientific International Inc.. The Grand Island location, also known as GIBCO, focuses on making cell culture media, which is essentially “food for cells” which is used for cell growth in research settings. There are two forms of cell culture media that is made at Thermo Fisher, Liquid and Powder. Each of these forms of cell culture media are divided by their material types, Animal Origin (AO) and Animal Origin Free (AOF). During my internship, I focused on Serum which is an AO liquid product.

Statement of Engineering Problem

Current routings for the sera product portfolio are outdated and inaccurate causing challenges for cost accounting and production planning. The scope of this project included creating a process map of the serum productions, performing time studies for process steps, creating an automated tool to generate the routing from the Bill of Materials and the Work Order listing.

Engineering Standards, Codes and Constraints

Codes:

- Clean Room gowning requirements(ISO 14698-1:2003)
 - If not gown certified, must be accompanied by someone who is
- Required to wear steel toes on the floor(ISO 20345:2021)

Constraints:

- Certain processes are run at specific times of the day
 - Sometimes I was required to come in at 2am to film time studies
- It is different jobs every day
 - Sometimes I was not able to see the job I wanted to time study due to it not being scheduled
- AO/AOF Segregation
 - Different colored scrubs must be worn in the respective areas and are not allowed to be worn in the other areas to avoid cross contamination

Engineering Approach and Analysis

In the development of the serum routings and the routing tool, there were several steps and engineering approaches that had to be made. First, a process map was made to understand how serum products are formed. Additionally, process maps were made of the specialty products to know how these products’ processes would change for the routings. Next, time studies were filmed and the technicians were interviewed to understand the process more deeply. The time studies were then used for when we would breakdown the time to determine how long each task took in the process. Then, from these time breakdowns we put them into the routing tool that is developed in excel and wrote formulas to make the tool automated.



Figure 1: 500 mL Bottles of Serum Product

Results

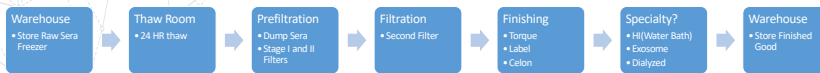


Figure 2: High Level Serum Process Map

Thermo Fisher Scientific							
Manufacturing Time Standard/Routing Matrix Using MOST							
MANUFACTURING - Central Dispensary							
Product SKU #:				ACQ:		1500 LT	
E1 ROUTING MATRIX							
ACQ *						1500	
SKU #	Work Center #	WC Name	Op Sequence	Run Labor	Setup Labor	Time Basis (Adjust for Run Labor)	Setup Crew
							1
							1
							1

Figure 3: Serum Routing Tool for Bulk SKUs

In figure 2, you will see the high level process map that was made to understand the serum production process. This was then used to develop the serum routing tool which is shown in figure 3. Figure 3 shows the serum routing tool for bulk SKUs and a similar one was made for all finished good SKUs. The way this tool works is that you will insert your SKU and ACQ in the respective locations and it will automatically generate the BOM of the product and the updated routing based on the ACQ, my data and the BOM. For one particular SKU, we found an 11 hour increase from the current to the future routing which means there was a \$3.57/liter increase from the current costing to the future costing. This will affect cost accounting with having accurate costing information and benefit SKU coordinators for over 100 serum SKUs.

Recommendations and/or Actions Taken

- Use data to make comparisons
 - When the new water baths were installed, we used the original data that I collected and compared it to the new data from the new water baths to show that time has been saved with the new water baths.
- Continuous Improvement Projects
 - Tool to improve ergonomics and standardize process in finishing when torquing the bottle caps
 - Decapper co-bot to assist with unscrewing the raw sera lids off their containers. This will improve ergonomics and save time during the process as it is a long and repetitive process.

References

- Thermo Fisher Scientific Inc.
- International Organization for Standardization. (2003). Cleanrooms and associated controlled environments — Biocontamination control — Part 1: General principles and methods (ISO Standard No. 14698-1:2003). <https://www.iso.org/standard/25015.html>
- International Organization for Standardization. (2021). Personal protective equipment — Safety footwear (ISO Standard No. 20345:2021). <https://www.iso.org/standard/73222.html>



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